

L07 — Chatbot III: Multimodality

Multimodal input, emotion recognition, personality

Study summary generated from lecture slides. ETH AI for Digital Characters.

Full interaction pipeline

- Speech-to-text -> feature recognition -> user state -> response generation
- -> chatbot state -> speech synthesis + animation
- User state: emotion, attention, age, gender, appearance, etc.

Emotion recognition

- Taxonomies: VAD (dimensional) vs basic emotions (categorical)
- Visual: landmarks -> CNN (VGG, EfficientNet) or ViT/Swin
- Audio: pitch, rhythm, intonation, energy from Mel spectrogram
- FACS for facial action units
- Fusion: early (concatenate features), late (fuse predictions), hybrid

Multimodal models

- CLIP: contrastive text-image pre-training, shared embedding space
- VLMs: joint visual + text processing (e.g. LLaVA)
- Personalization: user-specific fine-tuning on labeled data

Personality modeling

- Human Big Five (OCEAN): openness, conscientiousness, extraversion, agreeableness, neuroticism
- Chatbot traits: artificiality, conscientiousness, vibrancy, civility, neuroticism
- Dynamic Personality Infusion: trait intensity (-2 to +2) rewrites LLM response